OOP Basics

Overview

OOP stands for **Object-oriented programming.**

OOP is a style of programming where we build software using **objects** instead of just functions and data scattered around.

An **object** is something meaningful in your problem.

- For Example: user, bank account, order, employee, product, ticket, car.

Each object:

- **knows** something → its data (like name, balance, price).
- **does** something → its actions/behavior (like deposit, login, pay, print).

Problem with Procedural Programming

- **PoP** is series of instructions written in a **logical order**.
- For complex program, logic is broken down into **smaller blocks** of statements called **functions**.
- Returned value of one function can be used within other calling function(s).

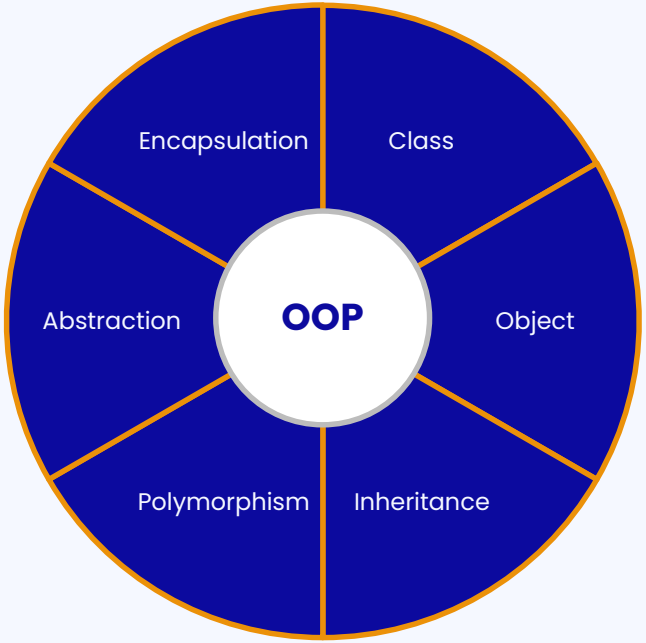
Problems:

- *Data and functions are **scattered***, so nothing clearly **"owns"** or protects important state.
- Heavy use of shared/global data *makes bugs **hard to track** and **easy to introduce***.
- **No clear boundaries or responsibilities**, so any part of the code can change anything.
- **Code becomes difficult to understand** and navigate as the project grows larger.
- Reusing and extending behavior often leads to **copy-paste code** and **inconsistencies**.

Why OOP?

- **Better Structure** – Groups related data and behavior into clear, self-contained units (objects).
- **Code Reuse** – Encourages reuse via classes and relationships instead of copy-paste.
- **Easy Maintenance** – Hides internals behind stable interfaces, so changes are safer.
- **Extensibility** – Allows adding new features/types without modifying existing core logic.
- **Real-World Modeling** – Represents real entities directly as objects, making designs intuitive.
- **Improved Testability** – Breaks systems into small, focused objects that are easier to test.

Characteristic of OOP



Class

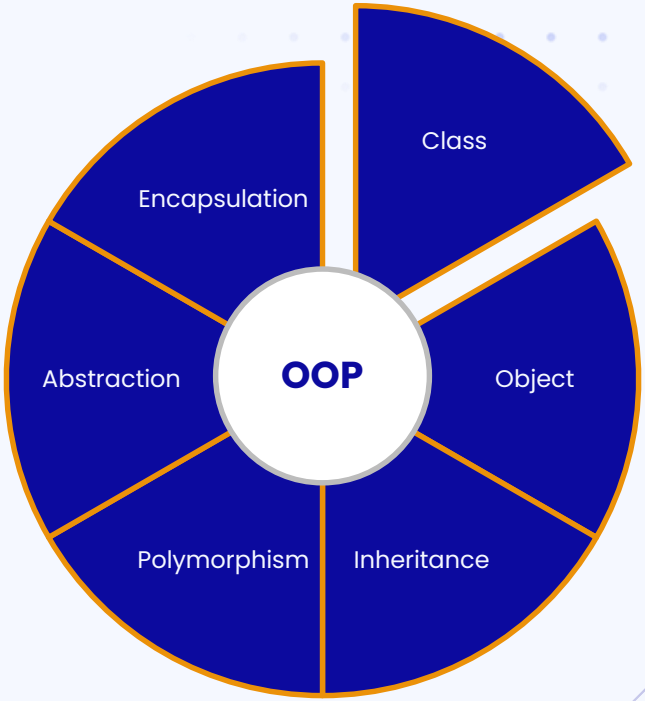
Definition:
A class is a blueprint or template that defines what a certain kind of object should know and do.

Structure:
It specifies the set of attributes (data) that its objects will hold.

Behavior:
It defines the operations/actions (methods) that its objects can perform.

Instance Creation:
Multiple individual objects (instances) are created based on one class.

Abstraction of a Concept :
It represents a general concept (e.g., “Student”, “Account”), not one specific example.



Object

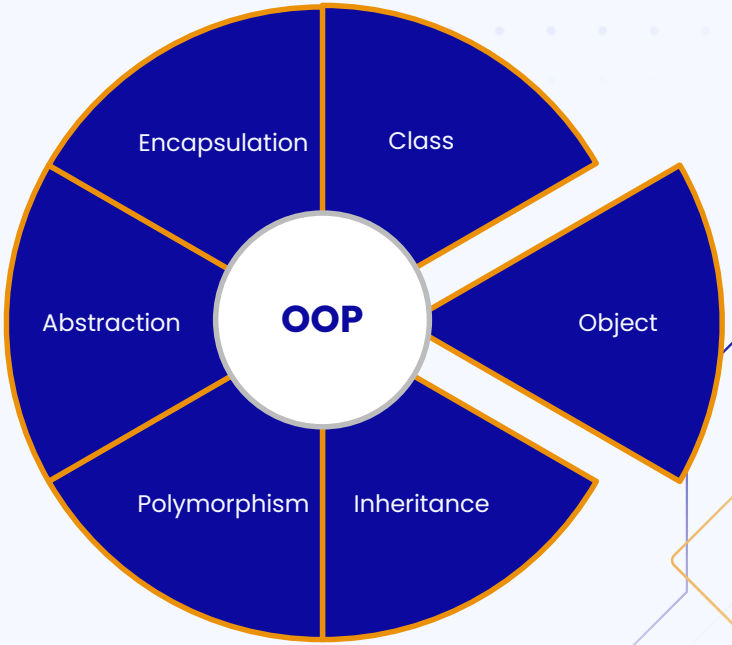
Definition:
An Object is an instance of a Class. It bundles related data and behavior together.

Instance of a Class:
It is a concrete example created from a class/blueprint.

State:
It stores its own data/values (like model, price, color), which can differ from other objects.

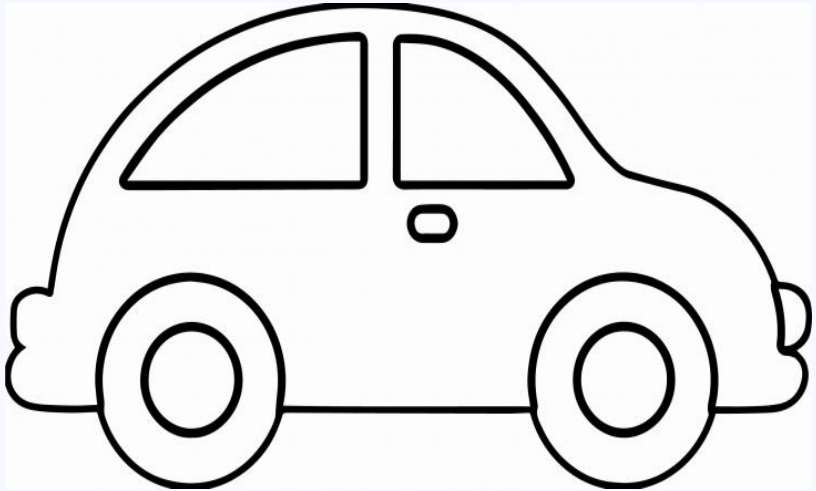
Behavior:
It can perform actions defined for its type (like accelerate, decelerate, break).

Identity:
Each object exists as a distinct unit, even if its data matches another object.



What is instance?

Instance = actual example or actual piece from a blueprint



Class (Generic Design / Blueprint)

Instances of Class (Real Specific Cars)

Inheritance

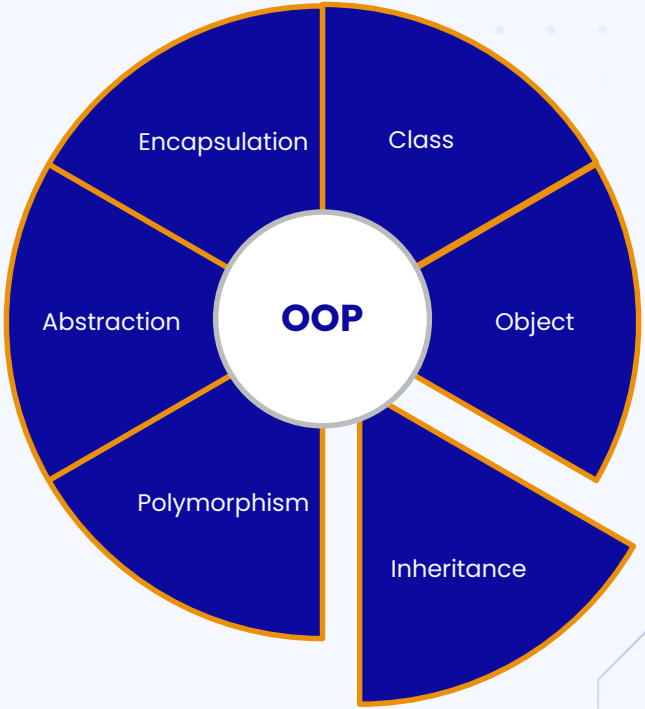
Definition:
Inheritance allows extending capability of an existing class (parent class) to build new class (child class)

Reuse:
Common attributes and behaviors are defined once in the parent and automatically available in the child.

Specialization:
Child classes can add new features or modify existing behavior to become more specific.

"Is-a" Relationship:
Inheritance models "X is a kind of Y" (e.g., Car is a kind of Automobile).

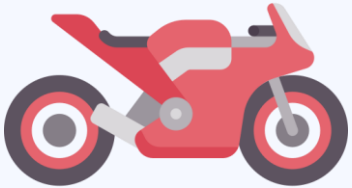
Less Duplication:
Reduces repeated code and keeps shared logic in one central place.



Inheritance Example

AUTOMOBILE

Car



Bike

Bus

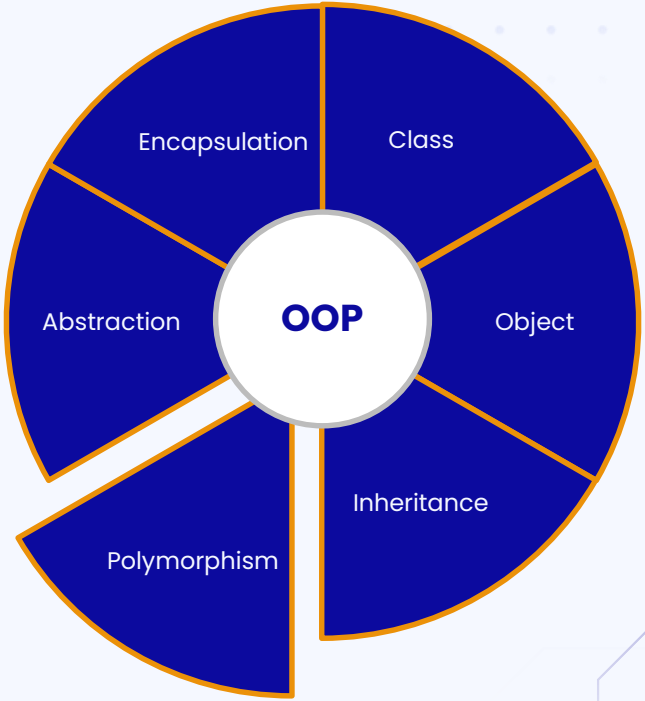
Polymorphism

Definition:
Polymorphism means the same operation or method name can work differently on different types of objects.

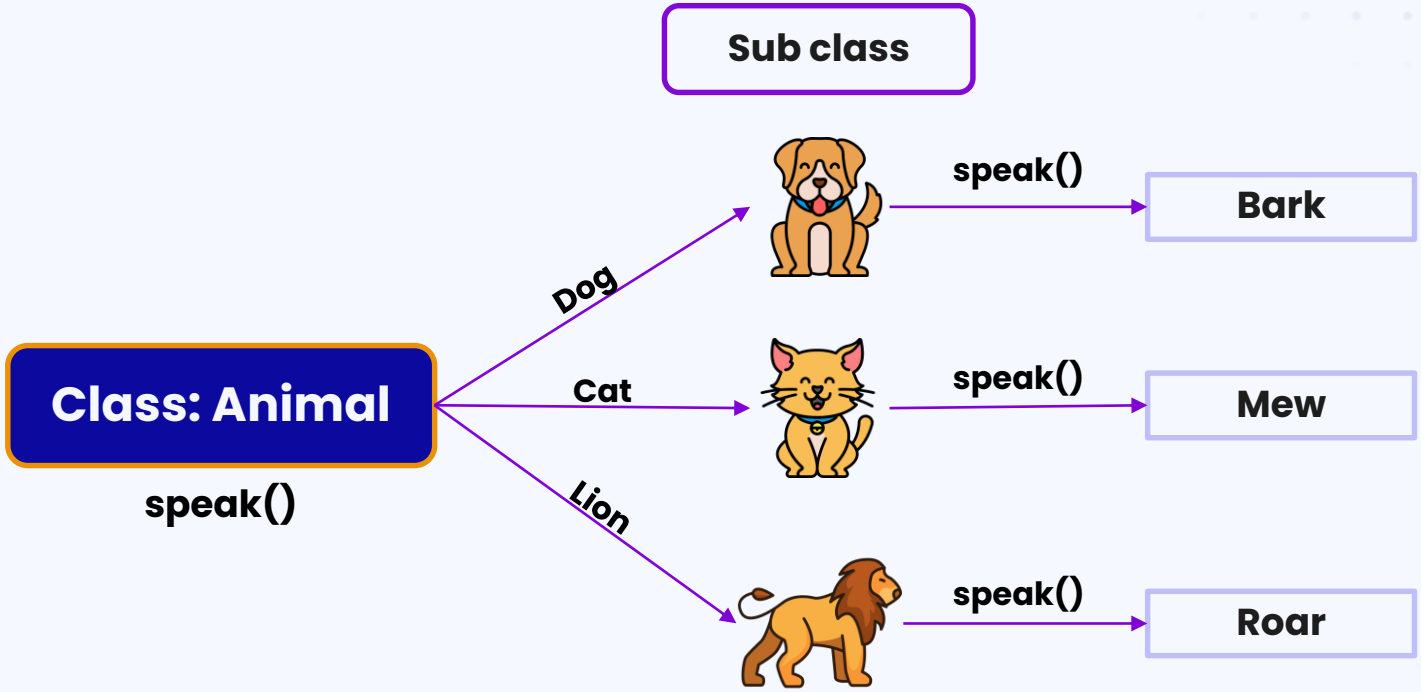
Common Interface:
Different classes provide the same method so they can be used interchangeably.

Different Behavior:
Each object gives its own implementation, but the calling code uses the same method call.

Supports Extensible Design:
Makes it easy to add new behaviors without modifying existing logic.



Polymorphism Example



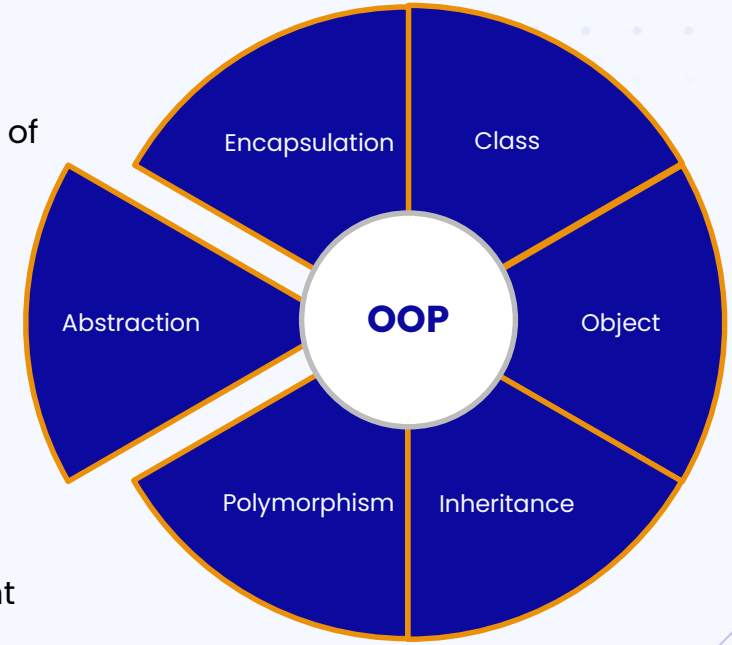
Abstraction

Definition:
Abstraction focuses on showing only the essential features of an object and hiding unnecessary details.

Reduced Complexity:
Helps users work with complex systems through simpler, high-level actions.

Implementation Hiding:
Internal logic can change without affecting code that depends on the exposed interface.

Foundation for Polymorphism:
Shared abstract behavior (common interface) lets different objects be used interchangeably.



Abstraction Example

We drive the car, not its engine internals.

Simple Controls Outside

Complex Logic Inside

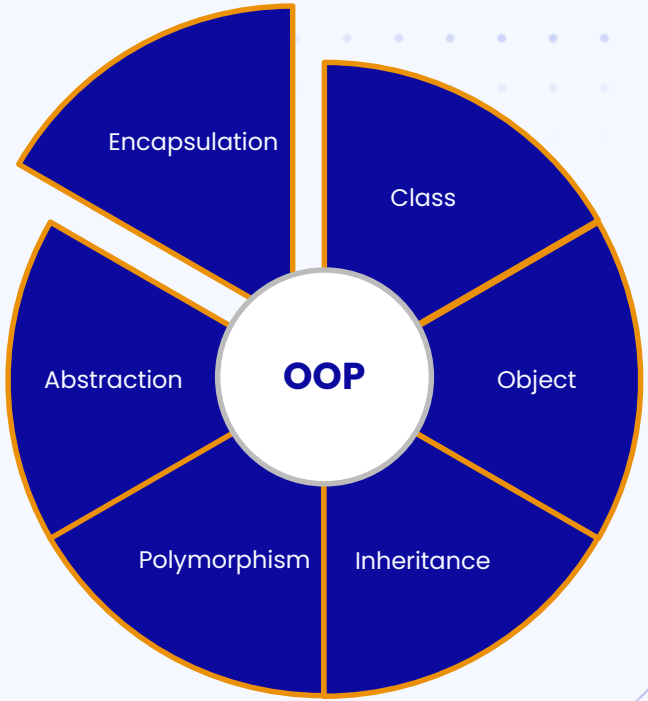
Encapsulation

Definition:
Encapsulation bundles data and the methods that operate on that data into a single unit (object).

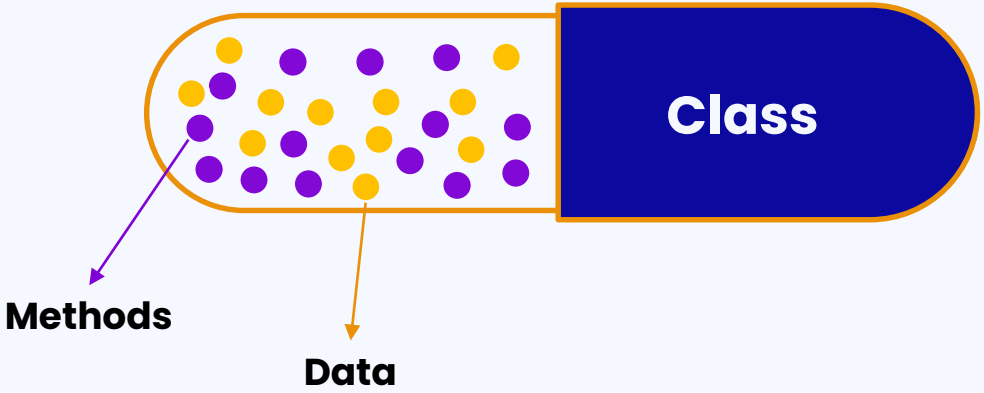
Controlled Access:
Internal details are accessed or changed only through well-defined methods, not directly.

Protection:
Prevents accidental or unauthorized modification of an object's internal state.

Change-Friendly:
Internal implementation can change without affecting code that uses the object's public interface.



Encapsulation Example



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Class & Object